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Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed.,
<https://www.probabilitycourse.com>

Hypothesis Testing – The Idea

In many situations, we want to decide whether a claim about the world is true — but we only have data from experiments or observations.

So we ask:

Does the data provide enough evidence to support a claim?

Two competing ideas:

- **Null hypothesis (H_0)**
The baseline or “default” assumption. It is assumed true unless data provide sufficient evidence against it.
- **Alternative hypothesis (H_1)**
The competing claim — what we accept if evidence contradicts H_0 .

Examples

Drug testing

We test a new drug on many patients.

- H_0 : The drug is **not effective**
- H_1 : The drug is **effective**

We collect data (e.g., patient outcomes) and check whether it would be *unlikely* if the drug had no effect.

Coin experiment

We flip a coin many times and record the results.

- H_0 : The coin is **fair** (probability of heads = 0.5)
- H_1 : The coin is **not fair**

If we observe too many heads (or tails), we may start to doubt H_0 .

Quiz

Each statement below describes a situation.

Choose the one **that cannot be formulated** as a pair of statistical hypotheses (H_0, H_1).

- A. A psychologist wants to test whether a new therapy reduces anxiety compared to no treatment.
- B. An engineer tests whether a new material is stronger than the standard one.
- C. A teacher wants to test whether her students score higher on a standardized test than students in other schools.
- D. A philosopher claims that beauty is subjective and cannot be measured statistically.

Choose a test Statistic

A random variable that summarizes the evidence about the hypothesis.

$$T = T(X_1, X_2, \dots, X_n)$$

Intuition

Instead of working with all the data, we extract something meaningful:

- Sample mean → tests about a mean
- Number of successes → tests about a probability
- Difference of means → compares two groups

The **test statistic** is the quantity we use to **decide whether to reject H_0**

Determine distribution under H_0

What we do

- Treat the data as generated under H_0
- Determine the distribution of T under H_0

$$T \mid H_0 \sim (\text{known distribution})$$

This distribution tells us:

What values of T are **typical** if H_0 is true

Typical outcomes

Depending on the problem, we may get:

- Exact distribution (e.g. binomial, Poisson)
- Approximate distribution (e.g. normal via CLT)
- Standardized form (e.g. Z , t , χ^2)

Make a statistical decision

Two equivalent approaches:

(A) Acceptance/rejection region approach

Rejection region

$$R = \{t : t \text{ is too unlikely under } H_0\}$$

such that:

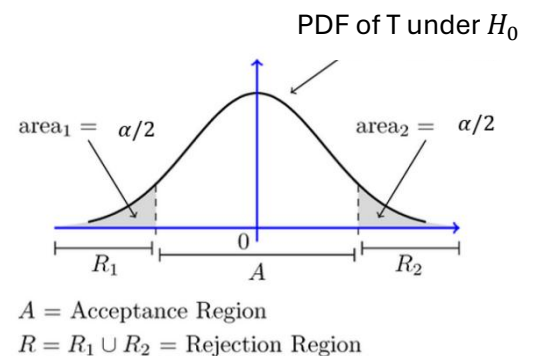
$$P(T \in R \mid H_0) = \alpha$$

Acceptance region

$$A = \{t : t \text{ is not too unlikely under } H_0\}$$

such that:

$$P(T \in A \mid H_0) = 1 - \alpha$$



(B) p-value approach

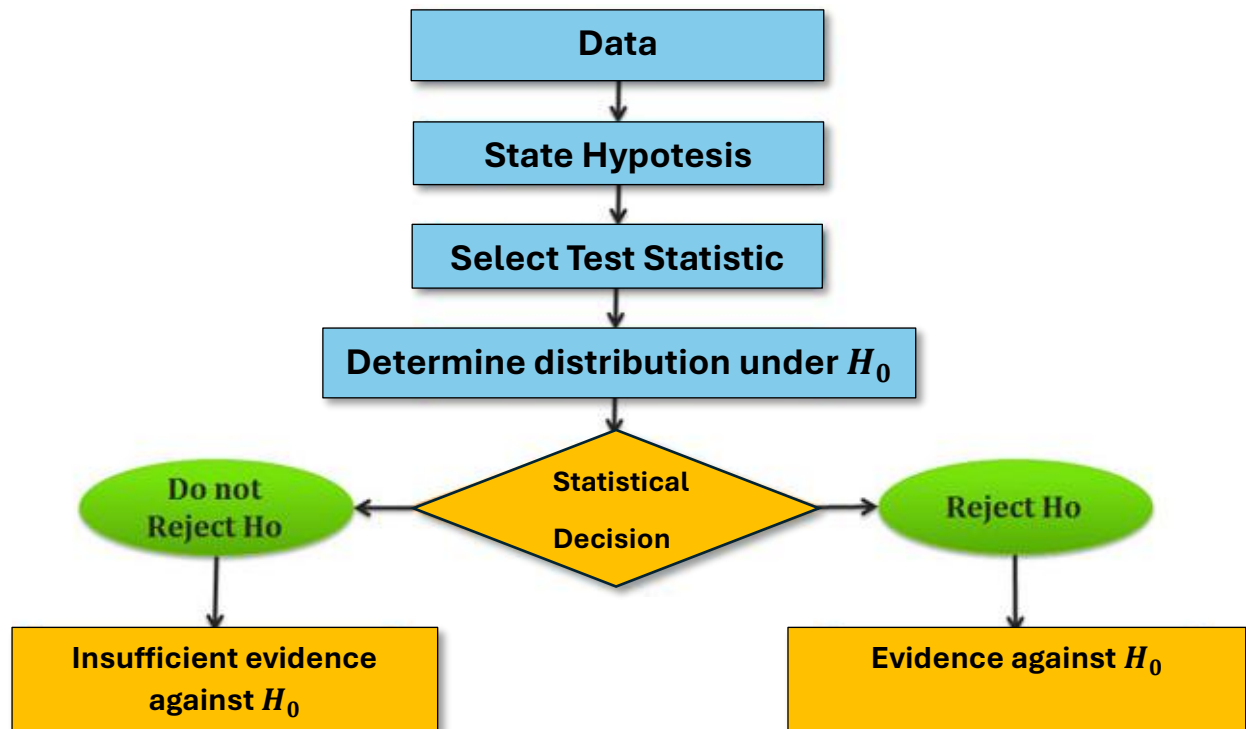
We compute:

$$p = P(\text{observing a value as extreme as } T_{\text{obs}} \mid H_0)$$

Then:

- If $p \leq \alpha \rightarrow$ reject H_0
- Otherwise $\rightarrow$ do not reject H_0

Steps in hypothesis testing



Example: Testing Whether a Coin is Fair**Step 1. Data**

We perform $n = 100$ independent tosses of a coin.

Let

$$X_i = \begin{cases} 1, & \text{if toss } i \text{ results in heads,} \\ 0, & \text{if toss } i \text{ results in tails.} \end{cases}$$

Assume the tosses are **independent and identically distributed (i.i.d.)**.

The total number of heads is

$$X = \sum_{i=1}^{100} X_i.$$

Step 2. State the Hypotheses

We let

$$\theta = P(\text{heads})$$

be the **true (but unknown)** probability of the coin landing on heads — our **parameter of interest**.

We want to test whether the coin is fair.

$$H_0 : \theta = \frac{1}{2} \quad (\text{the coin is fair})$$

$$H_1 : \theta \neq \frac{1}{2} \quad (\text{the coin is not fair})$$



Example: Testing Whether a Coin is Fair**Step 3. Select the Test Statistic**

Each $X_i \sim \text{Bernoulli}(\theta)$.

Thus,

$$X = \sum_{i=1}^{100} X_i \sim \text{Binomial}(100, \theta).$$

We use

$$X = \text{number of heads}$$

as our **test statistic**.

Under the null hypothesis H_0 .

Example: Testing Whether a Coin is Fair**Step 4. Determine the distribution under H_0**

Under the null hypothesis H_0 ,

$$X \mid H_0 \sim \text{Binomial}(100, \tfrac{1}{2}).$$

Expected value and variance under H_0 :

For a single coin toss $X_i \sim \text{Bernoulli}(p_0)$:

$$E[X_i] = \mu = p_0 = \tfrac{1}{2}, \quad \text{Var}(X_i) = \sigma^2 = p_0(1 - p_0) = \tfrac{1}{4}.$$

For the total number of heads

$$X = \sum_{i=1}^n X_i \sim \text{Binomial}(n, p_0),$$

we have

$$E[X] = n\mu = np_0 = 100 \times \tfrac{1}{2} = 50,$$

$$\text{Var}(X) = n\sigma^2 = np_0(1 - p_0) = 100 \times \tfrac{1}{2} \times \tfrac{1}{2} = 25,$$

so the standard deviation is

$$\sqrt{\text{Var}(X)} = \sigma\sqrt{n} = 5.$$

Example: Testing Whether a Coin is Fair**Normal Approximation (by the Central Limit Theorem)**

Nb. Instead of working directly with binomial tail probabilities, we standardize and use the standard normal distribution, which gives convenient critical values and simpler calculations.

By CLT we find:

$$Z = \frac{X - n\mu}{\sigma\sqrt{n}} = \frac{X - 100(\frac{1}{2})}{\sqrt{100(\frac{1}{4})}} = \frac{X - 50}{5} \approx \mathcal{N}(0, 1).$$

Example:

Suppose you perform the experiment:

- You toss the coin 100 times
- You observe, say:

$$X = 57$$

Then:

$$z_{\text{obs}} = \frac{57 - 50}{5} = 1.4$$

For each experiment (i.e., each dataset), the test statistic becomes a single constant.

Example: Testing Whether a Coin is Fair**Step 5. Statistical Decision****Approach A: Acceptance region**

We choose a rejection region R so that

$$P(T \in R \mid H_0) = \alpha.$$

In our case with a symmetric distribution and standardized test statistic Z under H_0 , we choose a critical value c such that

$$P(-c \leq Z \leq c \mid H_0) = 1 - \alpha.$$

Or equivalently

$$\Phi(c) - \Phi(-c) = 1 - \alpha \stackrel{\text{sym}}{\implies} \Phi(c) = 1 - \frac{\alpha}{2} \implies c = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$$

For $\alpha = 0.05$ we find:

$$c = \Phi^{-1}(0.975) = 1.96$$

Thus the rejection region is

$$R = \{z : |z| > 1.96\},$$

and the acceptance region is

$$A = \{z : |z| \leq 1.96\}.$$

Example: Testing Whether a Coin is Fair**Convert back to the scale of X**

$$-1.96 \leq \frac{X - 50}{5} \leq 1.96 \Rightarrow 50 - 9.8 \leq X \leq 50 + 9.8$$

(Rounded to Integers)

Since X counts the number of heads (an integer),

$$\begin{cases} 41 \leq X \leq 59 & \Rightarrow \text{Accept } H_0, \\ X \leq 40 \text{ or } X \geq 60 & \Rightarrow \text{Reject } H_0 \text{ (accept } H_1). \end{cases}$$

Interpretation

- Acceptance region: $A = \{41, 42, \dots, 59\}$
- Rejection region: $R = \{0, 1, \dots, 40\} \cup \{60, 61, \dots, 100\}$

Quiz

Take the coin test from before, and change the significance level to $\alpha = 0.01$.

Which set represents the **acceptance region**?

A. $\{43, \dots, 57\}$

B. $\{37, \dots, 63\}$

C. $\{38, \dots, 62\}$

D. $\{36, \dots, 64\}$

Example: Testing Whether a Coin is Fair**Approach B: p-value**

Instead of defining a rejection region in advance, we:

Compute how extreme the observed value is under H_0

For a **two-sided test**, "extreme" means:

$$|Z| \geq |z_{\text{obs}}|$$

So the p-value is defined as:

$$p = P(|Z| \geq |z_{\text{obs}}| \mid H_0)$$

Using:

$$|Z| \geq |z_{\text{obs}}| \iff Z \leq -|z_{\text{obs}}| \text{ or } Z \geq |z_{\text{obs}}|$$

So:

$$p = P(Z \leq -|z_{\text{obs}}| \mid H_0) + P(Z \geq |z_{\text{obs}}| \mid H_0)$$

or equivalently by symmetry

$$p = 2 P(Z \geq |z_{\text{obs}}| \mid H_0)$$

Express using the CDF

Recall:

$$P(Z \geq a) = 1 - \Phi(a)$$

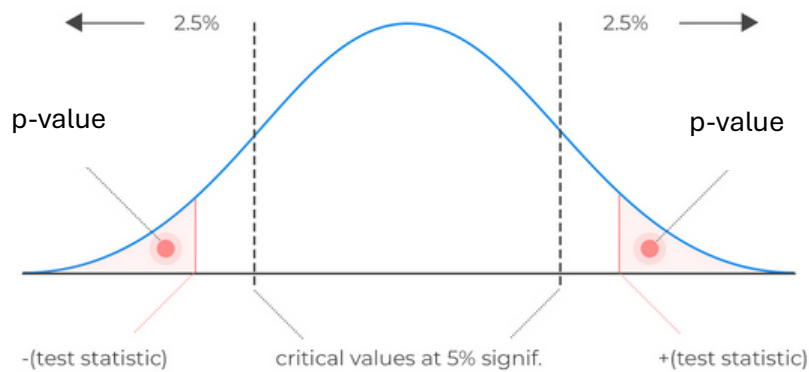
So:

$$p = 2(1 - \Phi(|z_{\text{obs}}|))$$

Example: Testing Whether a Coin is Fair**Decision rule**

We compare the p-value with α :

$$p \leq \alpha \implies \text{reject } H_0$$

**Connection to approach A**

we find the cutoff c such that:

$$p = \alpha$$

Substitute definition of p

$$2(1 - \Phi(c)) = \alpha$$

Solve for $\Phi(c)$

$$1 - \Phi(c) = \frac{\alpha}{2}$$

$$\Phi(c) = 1 - \frac{\alpha}{2}$$

Invert the CDF

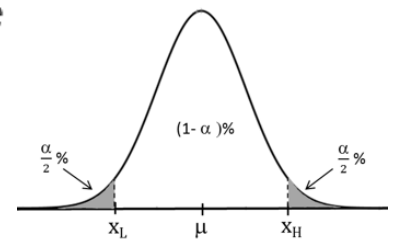
$$c = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$$

Types of tests

(a) Two-sided test

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0$$

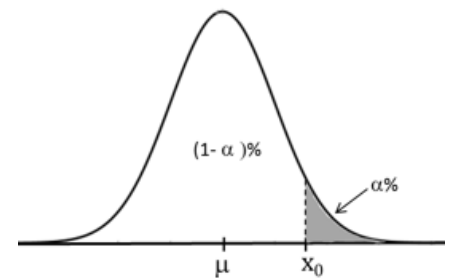
- H_0 : the parameter equals the hypothesized value
- H_1 : the parameter is **different** from θ_0



(b) Right-tailed (one-sided) test

$$H_0 : \theta \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0$$

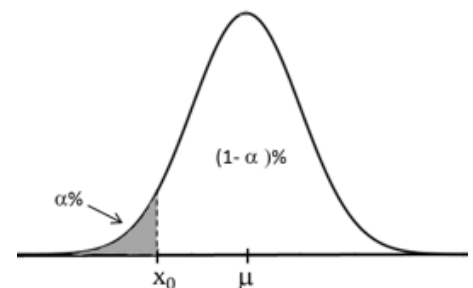
- Used when we suspect the parameter is **greater than** θ_0
- Reject when the test statistic is **too large**



(c) Left-tailed (one-sided) test

$$H_0 : \theta \geq \theta_0 \quad \text{vs.} \quad H_1 : \theta < \theta_0$$

- Used when we suspect the parameter is **less than** θ_0
- Reject when the test statistic is **too small**



General Setting and Definitions

Type I and Type II Errors

When we make a decision, two types of mistakes are possible:

- Type I Error:

$$P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ true}) = \alpha$$

→ The *significance level* of the test.

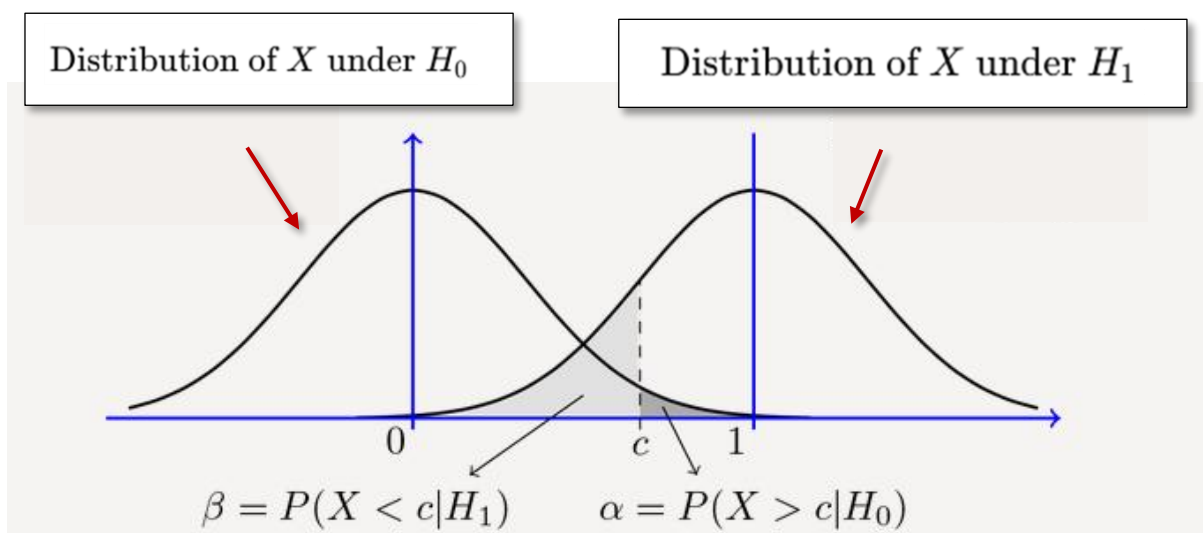
- Type II Error:

$$P(\text{Type II Error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ false}) = \beta$$

→ The probability of missing a false null hypothesis.

- Trade-off:

Decreasing the Type I error rate (α) generally increases the Type II error rate (β).



Example: Detecting Aircraft Using Hypothesis Testing

A radar system uses radio waves to detect whether an **aircraft** is present.

The system receives a signal X and must decide between two possibilities:

- **No aircraft is present**
- **An aircraft is present**

The radar system emits radio waves and listens for a **reflected signal**.

The received signal X is affected by two components:

$$X = (\text{signal from aircraft}) + (\text{noise from the environment})$$

Assume the noise is a particular normal distribution. Then:

$$X = \begin{cases} W, & \text{if no aircraft is present,} \\ 1 + W, & \text{if an aircraft is present,} \end{cases}$$

where $W \sim N(0, \sigma^2 = \frac{1}{9})$.

Thus, the model can be written compactly as:

$$X = \theta + W,$$

where

$$\theta = \begin{cases} 0, & \text{no aircraft (null hypothesis),} \\ 1, & \text{aircraft present (alternative hypothesis).} \end{cases}$$

Example: Detecting Aircraft Using Hypothesis Testing**Define the Hypotheses**

We define:

$$H_0 : \theta = 0 \quad (\text{no aircraft present})$$

$$H_1 : \theta = 1 \quad (\text{aircraft present})$$

Both H_0 and H_1 are **simple hypotheses** (each specifies a single parameter value).

- **Test statistic** is the observed signal X itself

We base our decision on the observed signal X .

- Under H_0 : $X \sim N(0, \frac{1}{9})$
- Under H_1 : $X \sim N(1, \frac{1}{9})$

We use a **threshold test**:

Reject H_0 if $X > c$.

This is a **one-sided test**:

We only reject for **large values** of X



Example: Detecting Aircraft Using Hypothesis Testing

Now that the hypotheses and the decision rule have been defined, we can analyze **how often the radar system makes errors** in its decisions.

 under H_0 , $X = W$
Type I Error Condition:

$$P(\text{Reject } H_0 \mid H_0) = P(X > c \mid H_0) = P(W > c) = \alpha$$

Standardization formula:

If $W \sim N(\mu, \sigma^2)$, then

$$Z = \frac{W - \mu}{\sigma} \sim N(0, 1).$$

Here, since $\mu = 0$ and $\sigma = \frac{1}{3}$,

$$Z = \frac{W - 0}{1/3} = 3W$$

Rewrite the probability

We want:

$$P(W > c)$$

Substitute $W = \frac{Z}{3}$:

$$P\left(\frac{Z}{3} > c\right) = P(Z > 3c).$$

Therefore,

$$P(Z > 3c) = 1 - P(Z \leq 3c) = 1 - \Phi(3c).$$

Example: Detecting Aircraft Using Hypothesis Testing

We want this probability to equal α :

$$1 - \Phi(3c) = \alpha.$$

Solving for c :

$$c = \frac{1}{3}\Phi^{-1}(1 - \alpha).$$

For $\alpha = 0.05$:

$$c = \frac{1}{3}\Phi^{-1}(0.95) = \frac{1}{3}(1.645) = 0.548.$$

Decision rule:

$$\text{Reject } H_0 \text{ if } X > 0.548.$$

Compute Type II Error Probability β

Type II error:

$$\beta = P(\text{Accept } H_0 \mid H_1) = P(X < c \mid H_1).$$

Under H_1 : $X = 1 + W$, so

$$P(X < c \mid H_1) = P(W < c - 1) = \Phi(3(c - 1)).$$

Substituting $c = 0.548$:

$$\beta = \Phi(3(0.548 - 1)) = \Phi(-1.356) = 0.088.$$

Thus, the probability of missing a present aircraft is 8.8%.

Hypothesis Testing for the Mean

Example: t-test

Average adult male height in the country is 170 cm. We suspect the mean in a specific city differs.

Random sample ($n = 9$) from that city (cm):

176.2, 157.9, 160.1, 180.9, 165.1, 167.2, 162.9, 155.7, 166.2

Assume heights are **normally distributed**.

Test at level $\alpha = 0.05$:

$$H_0 : \mu = 170 \quad \text{vs} \quad H_1 : \mu \neq 170$$

Determine the critical value

Because the **t-distribution** is symmetric about 0, we have:

$$P(T < -c) = P(T > c)$$

Hence, for a **two-sided test**, the total probability in both tails equals the significance level:

$$\alpha = 2 P(T > c)$$

or equivalently,

$$P(T > c) = \frac{\alpha}{2}.$$

Hypothesis Testing for the Mean

Example: t-test

Choose Significance Level

We choose $\alpha = 0.05$ for a two-tailed test.

Then:

$$P(T > c) = \frac{0.05}{2} = 0.025$$

Find c :

$$c = t_{1-\alpha/2, 8} = t_{0.975, 8} \approx 2.31$$

Thus, the acceptance region for the test statistic T is:

$$-2.31 \leq T \leq 2.31$$

Compute the Test Statistic

The test statistic for a one-sample t-test is:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Substitute the values:

$$T = \frac{165.8 - 170}{8.25/\sqrt{9}} = \frac{-4.2}{2.75} = -1.52$$

Hypothesis Testing for the Mean

Example: t-test

Decision Rule

Since $|T| = 1.52 < 2.31$, the test statistic is **not extreme enough** to reject H_0 .

Conclusion:

Fail to reject H_0 .

There is **no significant evidence** at the 5% level to conclude that the mean height in the city differs from 170 cm.

P-Values

Example: One-Sided Test for a Coin

We have a coin and want to test whether it is **fair**.

Let $\theta = P(\text{Head})$.

$$H_0 : \theta = 0.5, \quad H_1 : \theta > 0.5$$

We toss the coin **100 times** and observe **60 heads**.

We will:

1. Test H_0 at $\alpha = 0.05$
2. Test H_0 at $\alpha = 0.01$
3. Compute the **P-value**

P-Values

Example: One-Sided Test for a Coin

Define the test statistic

Let X = number of heads in 100 tosses.

Under H_0 :

$$E[X] = n\theta_0 = 50, \quad \text{Var}(X) = n\theta_0(1 - \theta_0) = 25.$$

Thus, approximately:

$$W = \frac{X - n\theta_0}{\sqrt{n\theta_0(1 - \theta_0)}} = \frac{X - 50}{5} \sim N(0, 1)$$

For our data ($X = 60$):

$$W = \frac{60 - 50}{5} = 2.$$

P-Values

Example: One-Sided Test for a Coin

Decision for given significance levels

(a) At $\alpha = 0.05$

Critical value:

$$c = z_{0.95} = 1.645$$

Decision rule: Reject H_0 if $W > 1.645$.

Since $W = 2 > 1.645$,

Reject H_0 .

(b) At $\alpha = 0.01$

Critical value:

$$c = z_{0.99} = 2.33$$

Decision rule: Reject H_0 if $W > 2.33$.

Since $W = 2 < 2.33$,

Fail to reject H_0 .

P-Values

Example: One-Sided Test for a Coin

Compute the P-value

The P-value is the *smallest* α for which we would reject H_0 .

For a **right-tailed** test:

$$P\text{-value} = P(W > w_1) = 1 - \Phi(w_1)$$

where $w_1 = 2$ is our observed statistic.

$$P = 1 - \Phi(2) = 1 - 0.9772 = 0.0228 \approx 0.023$$

- If we test at $\alpha = 0.05$, since $0.023 < 0.05$, we **reject** H_0 .
- If we test at $\alpha = 0.01$, since $0.023 > 0.01$, we **do not reject** H_0 .

Thus, the P-value = 0.023 tells us the **exact boundary** between rejecting and not rejecting.

P-Values

Example: Two-Sided Test for a Coin

We have a coin and want to test whether it is **fair**.

Let

$$\theta = P(\text{Head})$$

We test the hypotheses:

$$H_0 : \theta = 0.5, \quad H_1 : \theta \neq 0.5$$

We toss the coin **100 times** and observe **60 heads**.

We will:

1. Test H_0 at $\alpha = 0.05$
2. Test H_0 at $\alpha = 0.01$
3. Compute the **two-sided p-value**

P-Values

Example: Two-Sided Test for a Coin

Define the test statistic

Let X = number of heads in 100 tosses.

Under H_0 :

$$E[X] = n\theta_0 = 50, \quad \text{Var}(X) = n\theta_0(1 - \theta_0) = 25$$

Thus, approximately:

$$W = \frac{X - n\theta_0}{\sqrt{n\theta_0(1 - \theta_0)}} = \frac{X - 50}{5} \sim N(0, 1)$$

For our data ($X = 60$):

$$W = \frac{60 - 50}{5} = 2$$

P-Values

Example: Two-Sided Test for a Coin

Decision for given significance levels

(a) At $\alpha = 0.05$

For a **two-sided** test, we split the significance level equally between the two tails:

$$\frac{\alpha}{2} = 0.025$$

Critical value:

$$c = z_{1-\alpha/2} = z_{0.975} = 1.96$$

Decision rule:

Reject H_0 if $|W| > 1.96$.

Since $|W| = 2 > 1.96$,

Reject H_0 .

(b) At $\alpha = 0.01$

$$\frac{\alpha}{2} = 0.005, \quad c = z_{0.995} = 2.58$$

Decision rule:

Reject H_0 if $|W| > 2.58$.

Since $|W| = 2 < 2.58$,

Fail to reject H_0 .

P-Values

Example: Two-Sided Test for a Coin

Compute the P-value (Two-Sided Test)

The **P-value** is the smallest significance level α for which we would reject H_0 .

For a **two-tailed test**, we must consider *both tails* of the standard normal curve:

$$\text{P-value} = P(|W| > |w_1|) = 2P(W > |w_1|) = 2[1 - \Phi(|w_1|)]$$

where $w_1 = 2$ is our observed test statistic.

$$P = 2[1 - \Phi(2)] = 2(1 - 0.9772) = 2(0.0228) = 0.0456$$

- If we test at $\alpha = 0.05$, since $0.0456 < 0.05$, we **reject** H_0 .
- If we test at $\alpha = 0.01$, since $0.0456 > 0.01$, we **do not reject** H_0 .

Thus, the P-value = 0.0456 tells us that the result is just significant at the 5% level, but not significant at the 1% level.